A3. Premium Loyalty Discount & Late-Fee Ledger

Category B Assignment - Problem 3 (Intermediate)

Approach

The base class records every applied fee in a private fixed-size array. PremiumMember halves the amount and delegates to super.chargeLateFee(...). getLateFeeHistory returns a copy, preventing outside modification.

Java Solution

```
import java.util.Arrays;
class GymMember {
    protected final String memberId; protected final int monthlyFee;
    private final int[] lateFeeHistory = new int[10];
    private int feeCount;
    public GymMember(String id, int fee) {
        if (id == null || id.trim().length() < 4) throw new IllegalArgumentException("Invalid ID");
        if (fee <= 0) throw new IllegalArgumentException("Fee must be positive");
        memberId=id; monthlyFee=fee;
    }
    protected void chargeLateFee(int amount) {
        if (amount <= 0) throw new IllegalArgumentException("Fee must be positive");
        if (feeCount == lateFeeHistory.length) throw new IllegalStateException("Fee history full");
        lateFeeHistory[feeCount++] = amount;
    }
    public int[] getLateFeeHistory() { return Arrays.copyOf(lateFeeHistory, feeCount); }
    public int getTotalLateFees() {
        int total=0; for (int i=0;i<feeCount;i++) total+=lateFeeHistory[i]; return total;
    }
}
class PremiumMember extends GymMember {
    private final String trainerName;
    public PremiumMember(String id, int fee, String trainer) {
        super(id, fee); trainerName=trainer;
    }
    @Override protected void chargeLateFee(int amount) {
        super.chargeLateFee(amount / 2);
    }
}
class FeeLedgerDemo {
    public static void main(String[] args) {
        PremiumMember p = new PremiumMember("MEM5", 2000, "Coach Riya");
        p.chargeLateFee(200);
        int[] history = p.getLateFeeHistory(); history[0] = 999;
        System.out.println(p.getTotalLateFees());
        System.out.println(Arrays.toString(p.getLateFeeHistory()));
    }
}
```

Sample Output

```
100
[100]
```

Notes

The PDF keeps up to 10 fees per member as stated in the source.